R3.1 Proton transfer reactions

pH of salt solutions (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- construct equations for the hydrolysis of ions in a salt
- predict the effect of each ion on the pH of the salt solution.

Figure 1. Sodium hydrogencarbonate relieves heartburn caused by acid reflux.

Credit: Juanmonino, Getty Images

The salt sodium hydrogencarbonate is one of the ingredients in tablets that provide relief from heartburn. Can all salts be used to relieve heartburn? The salt sodium chloride, for example, does not give relief from heartburn. Why? In this section, you will learn that salts can be acidic, basic or neutral.

Salt hydrolysis

Neutralisation reactions were covered in section R3.1.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reactions-of-acids-id-46056/). Although salts are the products of a neutralisation reaction, they do not all form neutral solutions. Salt solutions can be acidic or basic too.

Salts of strong acids and strong bases

In section S2.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-of-ionic-compounds-id-43559/), you learned that ionic compounds will dissociate in water to form ions (**Figure 2**).

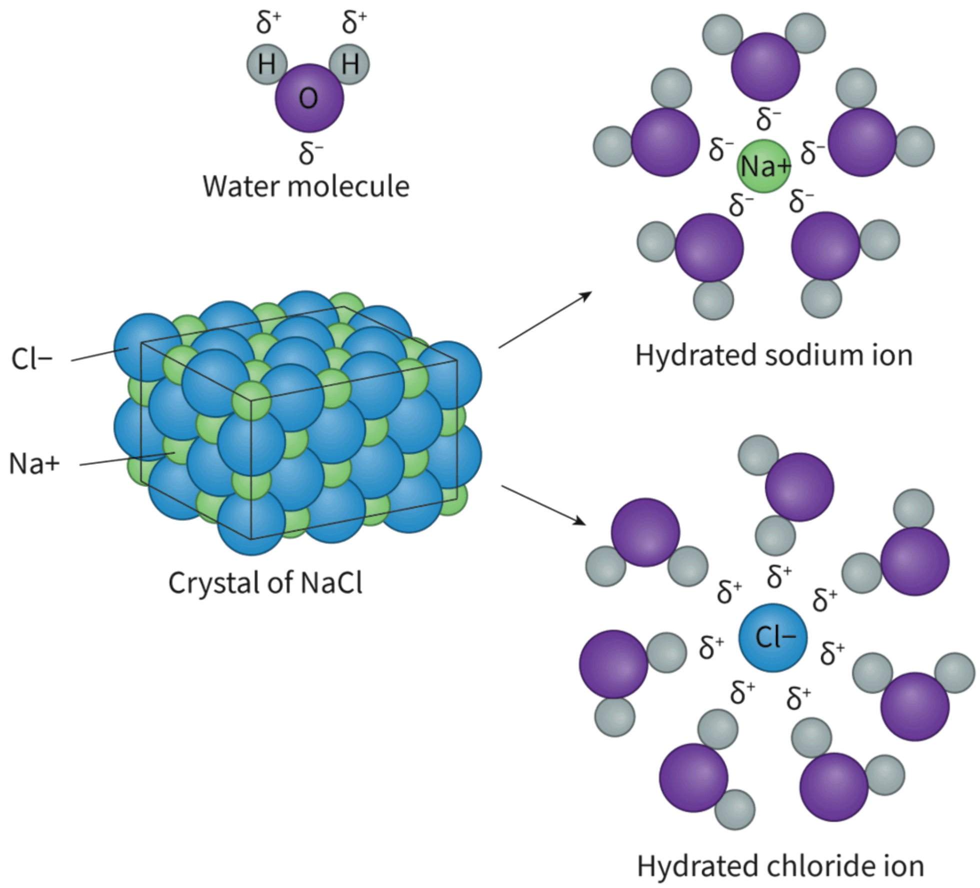

Figure 2. Dissolution of an ionic compound breaking apart the ions.

More information for figure 2

The diagram illustrates the process of sodium chloride (NaCl) dissociating in water to form hydrated ions. At the center left, there is a crystal structure of NaCl, depicted as a 3D lattice of alternating green spheres representing sodium ions (Na+) and blue spheres representing chloride ions (Cl−). Arrows point to two separate hydrated ions formations, showing the dissociation process.

At the top right, a sodium ion (Na+) is centrally surrounded by water molecules, depicted with purple and grey colors. Each water molecule has a partial negative charge (δ−) towards the sodium ion and a partial positive charge (δ+) on the hydrogen atoms.

Below, a chloride ion (Cl−) is surrounded by water molecules in a similar hydration shell, with the hydrogen end of each water molecule facing the chloride ion due to the attraction of opposite charges. The spheres surrounding both ions represent the hydration shells formed by the partial positive and negative charges of the water molecules.

[Generated by AI]

When a strong acid and strong base react, the resulting salt consists of ions that are neither acidic nor basic, which means that neither is able to hydrolyse water. The solution will, therefore, have a pH of 7 at 298 K. Examples of such salts are sodium chloride (NaCl), potassium nitrate (KNO_3) and sodium sulfate (Na_2SO_4). They dissociate to give neutral ions and thus a solution with a pH of 7.

$$NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$$

When sodium ions and chloride ions react with water, think about what possible products could form. Using ionisation equations, sodium cations could form a bond with hydroxide anions and chloride anions could form a bond with hydrogen cations, producing H^+ ions and OH^- ions, in their respective equations.

$$Na^+(aq) + H_2O(l) \rightleftharpoons NaOH(aq) + H^+(aq)$$

$$Cl^-(aq) + H_2O(l) \rightleftharpoons HCl(aq) + OH^-(aq)$$

Recall from section R3.1.6a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/), that sodium hydroxide is a strong base and hydrochloric acid is a strong acid. Strong acids and bases fully ionise in water, so therefore, you will not find these products in solution.

$$Na^+(aq) + H_2O(l) \leftarrow NaOH(aq) + H^+(aq)$$

$$Cl^-(aq) + H_2O(l) \leftarrow HCl(aq) + OH^-(aq)$$

Therefore, NaCl does not have any acid–base activity and will have a pH of 7 in solution.

Salts of weak acids and strong bases

An example of hydrolysis by the salt formed from a weak acid and strong base is that of sodium ethanoate (CH_3COONa). This salt fully dissociates in solution to form sodium ions and ethanoate ions:

$$CH_3COONa(s) \rightarrow Na^+(aq) + CH_3COO^-(aq)$$

As explained above, the sodium ion does not have acid-base activity. The ethanoate ion (CH_3COO^-) is the conjugate base reacts with water to form ethanoic acid, a weak acid, which does not fully ionise in water. This reaction will therefore show equilibrium, forming OH^- ions that make the solution alkaline, with a pH above 7 at 298 K.

Kog

$$CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$$

Salts of strong acids and weak bases

In the reaction of a strong acid and a weak base, the conjugate acid of the weak base is strong enough to hydrolyse water. For example, the reaction of hydrochloric acid and ammonia produces the ammonium ion according to the following reaction:

$$HCl(aq) + NH_3(aq) \rightarrow NH_4^+(aq) + Cl^-(aq)$$

As shown above, the chloride ion does not have acid–base activity. The ammonium ion reacts with water to form ammonia, a weak base, which does not fully ionise in water. This reaction will show equilibrium, forming H_3O^+ ions that make the solution acidic, with a pH below 7 at 298 K

$$NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$$

Salts of weak acids and weak bases

What do you think would happen to the solution if both of the ions in the salt showed acid–base activity?

The pH of a salt formed by the reaction of a weak acid and a weak base depends on the K_a and K_b values of the acids and bases involved. If the parent acid and base have approximately the same value, neither ion will hydrolyse and the salt produced will have a pH of around 7.0. Use **Interactive 1** to compete a summary of the pH of different salts produced in neutralisation reactions.

Kog

Reaction	Example	Product	Hydrolysis	Acidic, basic or neutral	pH at 298 K
Strong acid + strong base	hydrochloric acid + sodium hydroxide				
Weak acid + strong base	ethanoic acid + sodium hydroxide				
Strong acid + weak base	hydrochloric acid + ammonia				
Weak acid + weak base	ethanoic acid + ammonia				

ammonium ethanoate

ammonium chloride

sodium chloride

sodium ethanoate

ammonium ion hydrolyses water

neither ion hydrolyses water

both hydrolyse water

ethanoate ion hydrolyses water

acidic

basic

depends on strength of conjugate acid or base

neutral

cannot generalise

< 7.0

7.0

> 7.0

Check

Interactive 1. Summary table of the acidic and basic behaviour of salts.

Study skills

Examples should include the ammonium ion NH_4^+, the carboxylate ion $RCOO^-$, the carbonate ion CO_3^{2-}, and the hydrogencarbonate ion HCO_3^-.

The acidity of hydrated transition element ions and (aq) is not required.

Kog

Worked example 1

Predict if the pH is 7, less than 7, greater than 7 for the following salts in aqueous solution:

1. Ammonium chloride
2. Potassium ethanoate
3. Sodium hydrogencarbonate
4. Lithium carbonate
5. Lithium chloride

1.

Solution steps	Calculations
Step 1: Write dissociation equation.	$NH_4Cl(s) \rightarrow NH_4^+(aq) +$
Step 2: Identify which ion undergoes hydrolysis.	Ammonium ions hydrolyse
Step 3: Write the equation.	$NH_4{}^+(aq) + H_2O \rightleftharpoons NH$
Step 4: State the answer	The presence of hydroniur than 7.

2.

Solution steps	Calculations
Step 1: Write dissociation equation.	$CH_3COOK(s) \rightarrow K^+(aq) + CH_3CO$
Step 2: Identify which ion undergoes hydrolysis.	Ethanoate ions hydrolyse water.

Kog

Solution steps	Calculations
Step 3: Write the equation.	$CH_3COO^-(aq) + H_2O \rightleftharpoons CH_3COOH$
Step 4: State the answer	The presence of hydroxide ions makes the pH is more than 7.

3.

Solution steps	Calculations
Step 1: Write dissociation equation.	$NaHCO_3(s) \rightarrow Na^+(aq) + HC$
Step 2: Identify which ion undergoes hydrolysis.	Hydrogencarbonate ions hydro
Step 3: Write the equation.	$HCO_3{}^-(aq) + H_2O \rightleftharpoons H_2CO_3$
Step 4: State the answer	The presence of hydroxide ions therefore the pH is more than 7

4.

Solution steps	Calculations
Step 1: Write dissociation equation.	$Li_2CO_3(s) \rightarrow 2Li^+(aq) + CO_3{}^{2-}(a$
Step 2: Identify which ion undergoes hydrolysis.	Carbonate ions hydrolyse water.
Step 3: Write the equation.	$CO_3{}^{2-}(aq) + H_2O \rightleftharpoons HCO_3{}^-(aq)$

Kog

Solution steps	Calculations
Step 4: State the answer	The presence of hydroxide ions mak the pH is more than 7.

5.

Solution steps	Calculatic
Step 1: Write dissociation equation.	$\mathrm{LiCl(s)}$ –
Step 2: Identify which ion undergoes hydrolysis.	No hydrol
Step 3: State the answer.	Therefore

Worked example 2

Calculate the pH of 0.10 mol dm^{-3} of ammonium chloride (K_a of NH_4^+ = 5.6×10^{-10}).

Dissociation of salt:

$$NH_4Cl(s) \rightarrow NH_4^+(aq) + Cl^-(aq)$$

Therefore:

$[NH_4^+]$ = 0.1 mol dm^{-3}

Hydrolysis of ammonium, not chloride:

$$NH_4^+(aq) + H_2O \rightleftharpoons NH_3(aq) + H_3O^+(aq)$$

$$K_a = \frac{[NH_3][H_3O^+]}{[NH_4^+]}$$

As per previous K_a calculations, at equilibrium:

$[NH_3] = [H_3O^+] = x$

$[NH_4^+] = 0.10 - x = 0.10$
(remember the approximation since K_a is small)

$$K_a = 5.6 \times 10^{-10}$$
$$= \frac{x^2}{0.10}$$

Solving for *x*:

$x = 7.5 \times 10^{-6}$

Therefore:

$[H_3O^+] = 7.5 \times 10^{-6}$ mol dm^{-3}

pH = $-\log [H_3O^+]$
= 5.13
= 5.1 (2 s.f.)

Worked example 3

Calculate the pH of 0.10 mol dm^{-3} of potassium ethanoate (K_b of CH_3COO^- = 5.6 × 10^{-10}).

Dissociation of salt:

$$CH_3COOK(s) \rightarrow K^+(aq) + CH_3COO^-(aq)$$

Therefore:

$[CH_3COO^-] = 0.1$ mol dm^{-3}

Hydrolysis of ethanoate, not chloride:

$$CH_3COO^-(aq) + H_2O \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$$

$$K_b = \frac{[CH_3COOH]\,[OH^-]}{[CH_3COO^-]}$$

As per previous K_b calculations, at equilibrium:

$[CH_3COOH] = [OH^-] = x$

$[CH_3COO^-] = 0.10 - x = 0.10$
(remember the approximation since K_b is small)

$$K_a = 5.6 \times 10^{-10}$$
$$= \frac{x^2}{0.10}$$

Solving for x:

$x = 7.5 \times 10^{-6}$

Therefore:

$[OH^-]=7.5 \times 10^{-6}\ mol\ dm^{-3}$

$[OH^-] \times [H_3O^+] = 1.0 \times 10^{-14}\ mol^2\ dm^{-6}$

Therefore:

$[H_3O^+] = 1.3 \times 10^{-9}\ mol\ dm^{-3}$

$$\begin{aligned} pH &= -\log [H_3O^+] \\ &= 8.87 \\ &= 8.9\ (2\ s.f.) \end{aligned}$$

Worked example 4

Calculate the pH of 0.10 mol dm^{-3} lithium chloride.

Dissociation of salt:

$$LiCl(s) \rightarrow Li^+(aq) + Cl^-(aq)$$

No hydrolysis possible. Ions are neutral.

Therefore:

$pH = 7$

Activity

- **IB learner profile attribute:** Knowledgeable:
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Instructions

1. For the salts listed in the table, state the cation, the anion, the acidic/basic behaviour of the ions, and conclude about the acidic/basic behaviour of the salt.
2. Research the pH of these salts. Are your predictions correct?

Salt	Cation	Acidic? Basic? Neutral?	Anion	Acidic Neu
Ammonium nitrate				
Lithium fluoride				
Sodium carbonate				
Potassium ethanoate				
Lithium hydrogencarbonate				